

3.6 Numerical solution of equations

Pure Mathematics 3 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Newton-Raphson updates $x_{n+1} = x_n - f(x_n)/f'(x_n)$; if f' is small it may jump far or fail. Interval bisection gives a guaranteed bracket under continuity and sign change. Numerical answers require a reason for chosen rounding and a check at boundaries.

Worked example: follow the reasoning

For $f(x) = x^2 - 2$ and $x_0 = 1.5$, $x_1 = 1.5 - 0.25/3 \approx 1.4167$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 For $f(x) = x^2 - 3$ and $x_0 = 2$, calculate one Newton step.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$f(2)=1$ [1], $f'(2)=4$ [1]; $x_1=2-\frac{1}{4}=1.75$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.